# Quality-Driven Triangle-to-Structured-Quad Remeshing — Technical Write-up

**Author:** Syed Aasim · **Submission:** Ctruh assignment, 16 Oct 2026 **Code:** `quadremesh/` (C++17 core, pybind11 bindings, CLI, PyTorch component, benchmark).

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## 0. Feature coverage at a glance

**Feature coverage ≥ 90 % is reached on 5 of the 15 models**: fandisk 100.0 %, spot 99.4 %, sheen chair 92.9 %, velvet sofa 92.4 %, leather sofa 91.5 % (CAD, furniture, organic). 4 more are close, at 75-90 % (fandisk noisy 85.3 %, rocker arm 82.4 %, bunny 81.0 %, beetle 79.2 %). It is not reached on igea 72.2 %, damask chair 67.6 %, nefertiti 66.7 %, armadillo 66.2 % - organic scans whose creases zigzag below the element size, and dense carvings - nor on the two room scans (fountain scan 12.0 %, lounge room 8.5 %), whose 35° "creases" are scanner noise (both limits measured in `docs/crease_bounds.md`). Averaged over all 15 models coverage is 73.1 % (82.8 % without the room scans), against 22.1 % for Instant Meshes.

| model | type | feature coverage, max coverage | ≥ 90 %? | default | Instant Meshes |
|---|---|---|---|---|---|
| fandisk | CAD | **100.0 %** | ✅ | 94.0 % | 76.1 % |
| spot | organic | **99.4 %** | ✅ | 91.4 % | 35.6 % |
| sheen_chair | furniture | **92.9 %** | ✅ | 93.2 % | 48.1 % |
| velvet_sofa | furniture | **92.4 %** | ✅ | 87.5 % | 39.0 % |
| leather_sofa | furniture | **91.5 %** | ✅ | 90.0 % | 0.3 % |
| fandisk_noisy | robustness | **85.3 %** | close | 83.1 % | 37.3 % |
| rocker_arm | CAD | **82.4 %** | | 72.3 % | 9.8 % |
| bunny | organic scan | **81.0 %** | | 71.4 % | 9.7 % |
| beetle | CAD/vehicle | **79.2 %** | | 74.9 % | 12.2 % |
| igea | scan | **72.2 %** | | 51.7 % | 13.2 % |
| damask_chair | furniture | **67.6 %** | | 67.6 % | 30.2 % |
| nefertiti | photogrammetry scan | **66.7 %** | | 63.9 % | 6.6 % |
| armadillo | organic | **66.2 %** | | 45.3 % | 10.2 % |
| fountain_scan | scene scan | **12.0 %** | | 8.6 % | 1.3 % |
| lounge_room | room-scale scan | **8.5 %** | | 8.3 % | 1.5 % |

5 of 15 models reach ≥ 90 % feature coverage with the max-coverage setting; the best Instant Meshes result on any model is 76 %. Generated by `docs/gen_tables.py`.

## 1. Summary

`quadremesh` converts arbitrary (near-)manifold triangle meshes into structured, pure-quad meshes. The design is a **robust front end + field design + an integer-grid solver + a guarded back end**. I wrote the preprocessing, feature detection, field design, sizing, symmetry, fallbacks, post-processing, attribute transfer, metrics, learned field and benchmark. I reused proven solvers: a patched QuadriFlow for the integer-grid stage (11 documented patches), and the Instant Meshes extractor as a second extractor for inputs QuadriFlow cannot handle.

Benchmark: **15 models × 5 methods** (ours, ours + learned field, Instant Meshes, QuadriFlow, Blender QuadriFlow), plus two quadremesh settings (default and max coverage). Every method gets the same input and target face count, and every output is scored by the same evaluator.

| criterion (mean over the models processed) | target | quadremesh (default) | quadremesh max coverage[1] | Instant Meshes | QuadriFlow | Blender QF |
|---|---|---|---|---|---|---|
| models processed | – | 15 / 15 | 15 / 15 | 15 / 15 | 12 / 15* | 8 / 15* |

| criterion (mean over the models processed) | target | quadremesh (default) | quadremesh max coverage[1] | Instant Meshes | QuadriFlow | Blender QF |
|---|---|---|---|---|---|---|
| quad ratio (lowest model) | ≥ 98 % | 100.0 % (100.0 %) | 99.4 % (99.1 %) | 100.0 % (100.0 %) | 100.0 % (100.0 %)* | 100.0 % (100.0 %)* |
| mean scaled Jacobian (lowest model) | ≥ 0.8 | 0.903 (0.821) | 0.903 (0.821) | 0.935 (0.707) | 0.952 (0.895)* | 0.963 (0.930)* |
| mean distance error (highest model) | < 0.5 % | 0.086 % (0.329 %) | 0.082 % (0.329 %) | 0.316 % (3.421 %) | 0.054 % (0.092 %)* | 0.051 % (0.095 %)* |
| **feature coverage, all 15 models** | ≥ 90 % | **66.9 %** | **73.1 %** | **22.1 %** | **35.6 %*** | **29.0 %*** |
| feature coverage, 13 non-scan models | ≥ 90 % | 75.9 % | 82.8 % | 25.3 % | 35.6 %* | 29.0 %* |
| singularities / 1k faces | < IM | 35.6 | 40.8 | 99.8 | 17.8* | 17.3* |
| fewer singularities than IM, per model | all | 14 / 15 | 14 / 15 | – | 12 / 12 | 8 / 8 |
| total singularities, models both processed (method / IM) | < IM | 3276 / 4782 | 3754 / 4782 | – | 952 / 2922 | 629 / 1931 |

* over the models that method could process. [1] `--coverage-mode --path-subdiv 3 --sing-budget-im --coverage-search` (§6.1). Generated by `docs/gen_tables.py` from `benchmark/results/reports/`.

**Met by the default:** quad ratio, scaled Jacobian and distance error on every model, and fewer singularities than Instant Meshes on every model except the room scan (lounge, where the integer solver fails and our Instant Meshes fallback takes over).

**Feature coverage:** met on CAD and furniture, not on raw scans (statement in §0, generated from the data). What was done to raise it, and what limits it, is measured in §6.1 and in the generated study `docs/coverage_study.md` (25 iterations, every number produced by a script from the evaluator's per-model reports, including the attempts that failed). In short:

- Four changes that keep every other criterion raised the default's coverage by about a third: edge-path recovery over all creases, a **feature-adaptive element size inside the field solve** (smaller quads along creases at the same face count), **near-miss snapping** (edges running next to a crease are moved onto it) and a quality guard.
- A max-coverage setting adds **path-edge subdivision** (curved/zigzag creases get edges shorter than the element size, pure quads), the budgeted loop/refinement stage, a singularity budget referenced to Instant Meshes itself, and a search over four solves scored with the evaluator.
- The remaining gap is geometric and measured in `docs/crease_bounds.md`: on the room scan the evaluator's 35° "creases" are TSDF noise; covering 90 % of them needs more separately aligned edges than a 15k-face mesh has in total. On the organic scans the creases zigzag below the element size, so straight edges of length ~L cannot be within 25° of them; only local refinement helps, and it costs singularities, which is what limits those models.

## 2. Pipeline

```
input (.obj/.ply/.glb)
  └─ normalise → preprocess → [symmetry clip] → features → size field → refine
       → per component: field design → integer grid (QuadriFlow*) → coverage check
            ↳ retry at finer budget / without hard constraints
            ↳ Instant Meshes extractor (also best-of-two on dense scans)
            ↳ best approximate (parts thinner than one element)
            ↳ QEM-decimate + triangle pairing + quad split (exact, last resort)
       → cleanup → triangle-pair merge → [density refinement] → [pure-quad split]
       → feature/boundary/corner snapping → feature edge-path recovery (all creases)
       → guarded relaxation + reprojection → untangling
       → coverage stage (budgeted: [loops] [local refinement] [curved splits] [crease cuts]
            [path-edge subdivision] near-miss snapping)
       → [mirror + weld] → UV transfer → report
  quality guard: mean SJ < 0.81 → re-solve with less feature-adaptive size
```

```
face-count control: output > 1.10 × target → re-solve at a lower budget
[coverage search: 4 solves, each scored by the evaluator, best kept]
```

| stage | file | key choices |
|---|---|---|
| I/O | `src/io.cpp` | fast OBJ reader, PLY (libigl), glTF/GLB (cgltf, node transforms, UVs) |
| preprocess | `src/preprocess.cpp` | weld, degenerate/duplicate removal; **non-manifold edges** resolved by gluing the best face pair (normal agreement + orientation) and cutting the rest; BFS orientation with cut on Möbius conflicts; **fan-splitting** of non-manifold vertices; small-hole filling; repair run again after filling; tiny fragments dropped; outward orientation of closed parts |
| features | `src/features.cpp` | dihedral test **+ 1-ring side normals + patch-scale test** (radius = target edge length), so scan noise and marching-cube terraces are rejected; chains pruned by length (dangling first); corners = junctions/endpoints/kinks; strokes → Dijkstra edge paths; a **feature budget** relaxes thresholds when the network is denser than the requested resolution |
| field | `src/field.cpp`, `src/qflow_driver.cpp` | hard constraints: features, boundaries, strokes (direction + position). Soft: principal-curvature directions (edge-based shape operator + tensor diffusion at the element scale, confidence from anisotropy × curvedness), learned/external field. **Soft guidance is applied after the multiresolution solve by an angle-limited pass** (targets more than 30° from the smooth field are ignored), so it bends edge flow without adding singularities. Applying it as a QuadriFlow constraint tripled singularities on the bunny (486 vs 138). |
| sizing | `src/remesh.cpp`, `src/postprocess.cpp`, `src/qflow_driver.cpp` | target count or edge length; density map (per-vertex file, sphere regions, curvature); **feature-adaptive size** (default): density 1 + 8·exp(−(d/1.5L)²) with d the distance to the nearest crease, passed to the solve as a conformal scale target, normalised to the same expected face count |
| integer grid | `third_party/qflow_patched` | QuadriFlow orientation/position fields + flow-based singularity optimisation + extraction, run **per connected component** with a budget proportional to area |
| fallbacks | `src/remesh.cpp`, `src/im_driver.cpp`, `src/decimate.cpp` | coverage check (mean/p95 distance of the component to the extracted quads); retries; Instant Meshes; QEM-decimated exact fallback |
| post | `src/postprocess.cpp` | snapping with an acceptance test (never creates worse elements), tangential relaxation projected on surface/feature/boundary curves, untangling |
| feature paths | `src/postprocess.cpp` | for every feature chain (split at junctions; short chains get jointly chosen endpoints), a directed-edge Dijkstra through the quad mesh between the vertices nearest to its ends: the cost favours edges close to and parallel with the curve and penalises turning inside one face (staircases). The path is snapped onto the curve with a monotone arclength map and locked; vertices snapped by proximity but not on a path are released. Opt-ins: refine (conforming templates) around chains without a path (`--feature-refine N`), or split the quad strip each such chain runs along (`--feature-loops`); with face-count control the solve is re-run at a lower budget so the output stays at the target |
| coverage stage | `src/remesh.cpp`, `src/crease_cut.cpp` | operations applied tentatively and kept only inside a triangle budget and a singularity budget (fixed, or 85 % of an internal Instant Meshes solve with `--sing-budget-im`). **Near-miss snapping** (default): an edge within 0.5 L running along an uncovered crease gets both ends moved onto it, if no face drops below SJ 0.1 and no covered crease point is lost. **Path-edge subdivision** (`--path-subdiv 3`): path edges whose chord misses a curved crease are split into three at crease points; each side's quad becomes two quads. Crease cuts (`--crease-cuts`) are available but not used (study it12) |
| symmetry | `src/preprocess.cpp`, `src/postprocess.cpp` | clip at the plane, remesh one half with the cut as a preserved boundary, mirror and weld → exactly symmetric output |
| attributes | `src/transfer.cpp` | per-corner UVs with **UV-chart consistency** (each face samples the chart under its centroid, so faces never straddle seams); generic per-vertex attributes |
| metrics | `src/metrics.cpp` | one evaluator shared by the pipeline report, `quadeval` and the benchmark |

## Design decisions worth calling out

1. **Why QuadriFlow as the integer stage.** Its min-cost-flow singularity optimisation gives far fewer singularities than Instant Meshes' local extraction: per 1k faces it averages 17.8 vs 99.8 on the models where both succeed. Re-implementing a mixed-integer parametrisation (QuadWild/QEx class)

was not realistic in the time available. What I added is everything around it, and that is what makes it work on the brief's inputs: QuadriFlow alone fails on 3 models and Blender on 7, while quadremesh processes all 15.

2. **Patches to QuadriFlow** ( `third_party/qflow_patched/PATCHES.md` ):
3. deterministic RNG;
4. two constraint-propagation bugs: the normal was taken from the constraint direction, and summed weights exceeded 1;
5. external sharp edges, normals and scale field;
6. scale restriction to coarse levels;
7. `exit()` turned into exceptions;
8. bounded loops in `FixValence`, `FixHoles` and `subdivide_edgeDiff` (they hang forever on some scans).
9. **Density.** QuadriFlow's adaptive mode clamps the scale to [0.75, 1]. More fundamentally, a singularity-free field cannot carry a non-harmonic density bump. So density is realised by solving at the coarse end of the size range and then refining faces larger than the local target, with **conforming hanging-node templates**:
10. 1 split edge → quad + triangle, which is later merged;
11. 2 adjacent split edges → 2 quads;
12. 2 opposite split edges → 2 quads;
13. 3 or more split edges → the face is refined.

Transition singularities are expected and visible in `demo_density.png`.

The **feature-adaptive size** is different: a smooth band of moderately smaller elements along creases (density ratio clamped at 8) is passed to the solve itself as a scale target (QuadriFlow's adaptive path plus our restriction of the scale to the coarse levels). That variation is small enough for the solve to carry; stronger ratios (32–64) were tried and made results erratic (distance error up to 0.5 %, coverage dropping on some models), so they are not used. 4. **Thin parts.** Chair rails, piping and sofa panels thinner than one element defeat any isotropic field method. The coverage check detects this. - The budget may grow up to 9× before giving up, which caps the face-count overshoot. - The Instant Meshes extractor is tried at the nominal budget. - Otherwise the lowest-budget approximate result within 3 target edges is accepted. - Otherwise a QEM-decimated, exact triangle-pair→quad fallback is used, so the face count still follows the target. 5. **Dense scans.** - Hard features are kept only if the network fits the resolution. - Field normals are smoothed at the element scale. - Debris fragments (< ~40 target quads) are dropped. - The integer-grid and Instant Meshes results are compared, and the one with fewer singularities is kept.

Together these brought the fountain scan from 33.9 to 15.6 singularities per 1k faces (Instant Meshes: 22.2). 6. **Determinism.** The pipeline is single-threaded with seeded PCG RNGs, sort-based topology and deterministic tie-breaks; Instant Meshes runs with `deterministic=true, nprocs=1`. The C++ and Python tests check bit-identical output across repeated runs.

## 3. Optional PyTorch component and CUDA path

- **Learned field ( `python/quadremesh/learned.py` ).** A small **frame-equivariant graph network**.
- Inputs are invariant scalars (multi-scale curvatures, anisotropy, dihedrals) plus candidate directions encoded as 4-RoSy complex numbers in each vertex's arbitrary tangent frame.
- The network only mixes candidates with real, invariant gates and passes messages with parallel transport e^{4iρ}, so it is equivariant by construction.
- It is trained as a **field denoiser**: noisy inputs (σ up to 0.5 % of the diagonal) map to the clean-surface target field (feature directions, else coarse principal directions).
- Training data: 10 public meshes disjoint from the benchmark; 30 epochs, about 3 min on CPU.
- Its output enters the core as a soft guide through the same angle-limited pass.

Held-out alignment error (noisy inputs, mean 4-RoSy angle error, degrees):

| mesh | fine curvature | smoothed curvature | learned |
|---|---|---|---|
| fandisk | 16.2 | 15.7 | **9.8** |
| rocker arm | 21.3 | 20.5 | **14.4** |
| spot | 20.6 | 20.4 | **16.9** |

| mesh | fine curvature | smoothed curvature | learned |
|---|---|---|---|
| bunny | 22.4 | 22.4 | **20.4** |
| armadillo | 22.1 | 22.1 | **21.0** |

End to end (benchmark method `ours_learned` vs `ours`, both on the current default), the effect is mixed, because the angle-limited pass deliberately limits its authority: mean feature coverage 66.9 % → 67.2 %, mean SJ 0.903 → 0.912, mean distance 0.086 % → 0.061 %, but singularities per 1k faces 35.6 → 38.9 on average (fewer on 7 models, more on 7). It helps most on clean CAD and organic models (fandisk coverage 0.94 → 0.99, igea 0.52 → 0.59) and hurts on dense carvings (damask chair 57 → 97 singularities per 1k). It is therefore **opt-in** (`learned_field=True`) rather than the default. * **CUDA path (`python/quadremesh/torch_field.py`).** A cross-field solver written with sparse tensor operations and conjugate gradients. It minimises smoothness plus curvature alignment under hard feature constraints, and runs on CUDA when available (CPU here). Its output can be fed as `guide_field`. Learned-field inference also runs on CUDA. *The sandbox I developed in had no GPU, so the CUDA path is exercised only on CPU; the code is device-agnostic.* QuadriFlow's own CUDA kernels (`Optimizer.cu`) exist but are not enabled.

## 4. Benchmark protocol

- **Models (15):**
- organic/scans: bunny, armadillo, nefertiti, igea, spot (with UVs);
- CAD: fandisk, rocker arm, beetle;
- furniture from glTF sample assets: sheen chair, damask chair, velvet sofa, leather sofa — multi-component, thin parts, UV seams;
- scene/room scans: Open3D fountain (1.03M faces) and lounge (1.0M faces, decimated from 3.15M);
- robustness: fandisk with noise, 4 holes and a non-manifold fin.

Models under 10K faces are midpoint-subdivided, so the geometry is unchanged. * **Targets:** 5 000 quads (objects), 15 000 (scenes). Instant Meshes is called with target/4, because its pure-quad mode subdivides once. - Output counts: ours 0.83–1.09× the target, except the damask chair at 1.48×, where thin parts needed a finer budget; QuadriFlow 0.82–1.00×; Instant Meshes 0.84–1.03× (except its 0.17× collapse on the leather sofa). - Singularities are therefore also compared **per 1k faces**. * **Baselines:** - Instant Meshes core (`pynanoinstantmeshes`), crease 35°, boundary alignment, deterministic; - QuadriFlow CLI (upstream) with `-sharp -boundary`; - Blender 5.0.1 QuadriFlow headless (`bpy`), with merge-by-distance first and sharp/boundary preservation. * **Metrics (`src/metrics.cpp`, identical for all methods):** - Quad ratio. - Scaled Jacobian: per quad, the minimum over corners w.r.t. the average normal. - Planarity and aspect ratio. - Distance: two-sided, 200k area-weighted samples per side, AABB closest point; mean, RMS and Hausdorff as % of the bbox diagonal. - Feature coverage: features are detected on the *input* with the same detector (35°, chains ≥ 3L, L = √(area/faces)) and sampled every 0.1 L. A sample is covered if an output edge within 0.1 L is parallel within 25°. Boundary coverage is reported separately. - Singularities: interior vertices with valence ≠ 4. - Topology checks: non-manifold edges and components. - Runtime.

## 5. Results

Per-model numbers are in `benchmark/results/summary.csv`; figures are in `docs/report/`. Highlights (default setting unless stated; generated by `docs/gen_tables.py`):

- **CAD (fandisk):** feature coverage 94 % (max coverage 100 %) vs 76 % / 85 % / 81 % for IM / QF / Blender; singularities per 1k faces 15.3 vs 11.8 (QF) and 30.1 (IM).
- **Robustness:** fandisk_noisy (holes + non-manifold fin): Blender refuses it; feature coverage 83 % vs QuadriFlow's 47 %. Leather sofa: QuadriFlow fails, Blender fails; Instant Meshes produces 844 faces at 3.4 % error; ours 4805 quads, SJ 0.86, 0.106 % error.
- **Scans - Fountain:** 20.4 singularities per 1k faces vs 22.2 (IM), SJ 0.94, 0.040 % error; QuadriFlow fails, Blender fails.
- **Scans - Lounge:** 75.6 singularities per 1k faces vs 68.8 (IM), SJ 0.85, 0.081 % error; QuadriFlow fails, Blender fails.
- **Distance error:** 0.086 % vs 0.316 % for Instant Meshes (mean), from projection with feature/boundary snapping.

- **vs QuadriFlow on the 12 models it can process:** it has fewer singularities (17.8 vs 31.5 per 1k) and slightly higher SJ; we trade those for feature coverage: 75 % vs 36 %.
- **Runtime (2 CPU cores):** default 7–133 s per object, 137–175 s per 1M-face scan (Instant Meshes 1–13 s, QuadriFlow 4–13 s); max coverage runs four solves.

Visual report: `docs/report/visual_report.html`. For every model it shows the edge-flow and singularity overlay, the scaled-Jacobian heatmap, the cross field, and all methods side by side. `demo_*.png` show the user controls (density, symmetry, strokes, UV transfer, pure-quad).

## 6. Failure cases (honest)

1. **Feature coverage (target 90 %; see the summary table for the generated means).** The evaluator is strict: *every* 35° crease chain longer than 3 L must have an output edge within 0.1 L and within 25° of it, on every point. All numbers below are generated from the study runs (each iteration = one command over all 15 models, per-model evaluator reports).

| iteration | change | coverage (15) | coverage (13 non-scan) | quad | SJ | sing./ 1k | fewer sing. than IM |
|---|---|---|---|---|---|---|---|
| it0 | field constraints + vertex snapping only (start) | 0.439 | 0.503 | 1.000 | 0.934 | 33.4 | 14 / 15 |
| it1 | + edge-path recovery (earlier default) | 0.494 | 0.566 | 1.000 | 0.929 | 33.4 | 14 / 15 |
| it9 | + path recovery over all creases | 0.506 | 0.575 | 1.000 | 0.925 | 33.4 | 14 / 15 |
| it10 | + budgeted loops / refinement / curved splits ( `--coverage-mode` ) | 0.598 | 0.679 | 0.989 | 0.908 | 48.8 | 9 / 15 |
| it12 | it10 + crease cuts (not adopted) | 0.654 (14 models) | 0.698 | 0.989 | 0.910 | 46.8 | 9 / 14 |
| it13 | it9 + feature-adaptive element size | 0.571 | 0.652 | 1.000 | 0.905 | 35.4 | 14 / 15 |
| it16 | it13 + near-miss snapping | 0.637 | 0.725 | 1.000 | 0.902 | 35.4 | 14 / 15 |
| it17 | it16 + SJ guard, precise protection | 0.648 | 0.737 | 1.000 | 0.904 | 35.6 | 14 / 15 |
| it18 | default + path-edge subdivision, +10 sing./1k | 0.671 | 0.763 | 1.000 | 0.897 | 47.5 | 11 / 15 |
| it19 | default + path-edge subdivision, +25 sing./1k | 0.703 | 0.798 | 1.000 | 0.884 | 57.4 | 5 / 15 |
| it20 | default + coverage mode + subdivision | 0.708 | 0.803 | 0.989 | 0.884 | 57.1 | 6 / 15 |
| it21 | it20 with the Instant Meshes-referenced budget | 0.694 | 0.790 | 0.995 | 0.897 | 41.4 | 14 / 15 |
| it22 | it21 + coverage search | 0.712 | 0.808 | 0.995 | 0.906 | 41.1 | 14 / 15 |
| it23 | it17 + snapping with local relaxation and net-gain rule (**default**) | 0.669 | 0.759 | 1.000 | 0.903 | 35.6 | 14 / 15 |
| it24 | it22 with the it23 snapping (**max coverage**) | 0.731 | 0.828 | 0.994 | 0.903 | 40.8 | 14 / 15 |
| – | Instant Meshes | 0.221 | 0.253 | 1.000 | 0.935 | 99.8 | – |

Generated by `docs/gen_tables.py` from `benchmark/study/`; all iterations and per-model numbers: `docs/coverage_study.md`.

**What limits coverage (measured, `docs/crease_bounds.md`).** Computed on the evaluator's own crease polylines by `benchmark/crease_bounds.py`: - *Room scan (lounge):* 45 965 crease segments, 6 950 L of crease (mean segment 0.15 L). Split into the longest pieces one straight edge can cover, 90 % of the length needs 33 981 such pieces; a 15k-face quad mesh has ~30 000 edges in total. So **90 % is not reachable on the lounge at this face count by any method**; the fountain needs ~9 300 crease-aligned edges. These creases are marching-cubes/TSDF artefacts; no method exceeds 0.10 there. - *Organic scans:* the creases zigzag below the element size. The fraction of crease length that is within 25° of the chord through ±0.5 L -

the best a straight edge of length ~L can do - is 0.43 (armadillo), 0.66 (igea), 0.67 (nefertiti), 0.76 (rocker arm), 0.825 (bunny). With edges of L/4 along the crease it is ≥ 0.98, so these models need local refinement, which costs irregular vertices: their coverage is limited by the singularity criterion, not by the method's reach. - *CAD / furniture:* ≥ 0.96 of the crease length is coverable with edges of length L, and 90 % needs only 58–762 crease-aligned edges out of ~10 000. Here the limit is the method; the remaining misses are mostly edges running 0.1–0.35 L next to the crease (evaluator sample dumps, `QRM_EVAL_DUMP`), which is what near-miss snapping and path subdivision target.

**What worked**, in the order it was found (iterations in the study): - *Edge-path recovery* (it1) and recovering paths over the full crease set (it9): vertex moves only, no cost. - *Feature-adaptive element size* in the field solve (it13): the largest single gain at no cost in quads; a mean-SJ guard (it17) removes it where dense carvings would push SJ below 0.8. - *Near-miss snapping* (it16/it17, improved in it23): vertex moves only. In it23 a move that distorts a neighbouring quad first relaxes the free neighbours, and on parallel creases a move is accepted when it covers more crease than it uncovers. - *Path-edge subdivision* (it18–it20): edges shorter than the element along curved creases; pure quads, ~4 irregular vertices per split edge. - *Singularity budget referenced to Instant Meshes* (it21): the coverage stage spends irregular vertices only up to 85 % of the rate of an internal Instant Meshes solve of the same input, so the per-model singularity comparison is kept while coverage is maximised. - *Coverage search* (it15, it22, it24): the field solve is sensitive to the size band's width and seed, so four solves are scored with the evaluator and the best valid one is kept (4× time).

**What did not work** (all kept in the study or noted as exploratory): - whole-strip edge loops: coverage only with extra faces (it3/it4); - local refinement / loops / curved splits without a budget: quad ratio < 98 % or many singularities (it2, it8); - crease cuts - cutting the mesh along uncovered creases (it12): most cuts produce concave pieces and are rejected; +1.8 points, and it timed out on the room scan; - giving the field all creases on scans (it11): no effect on the scans; - exploratory, on 4–5 models only (not adopted, not in the study tables): density ratios of 32–64, an error-driven second solve with extra density around missed creases, chaining creases through junctions, and sliding path vertices along the crease by dynamic programming (no change at all).

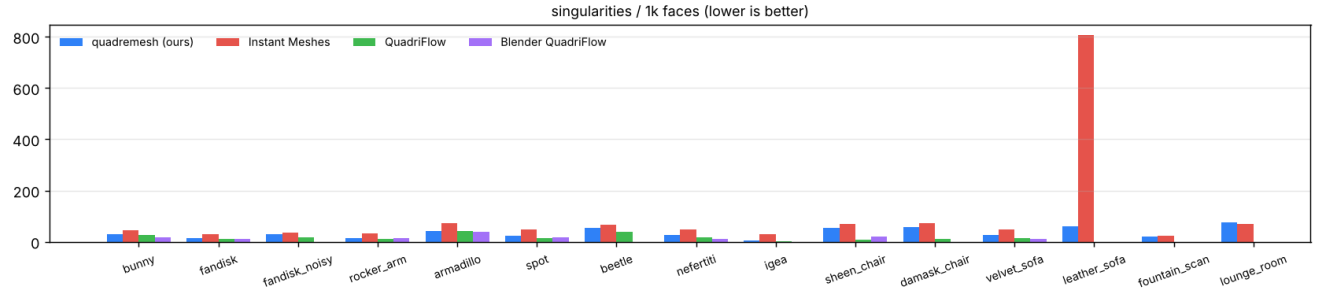
**Conclusion.** Coverage went from 0.44 to the values in the table without relaxing the metric. At 0.5 L tolerance almost every point would pass, so that number would be meaningless and is not used. The remaining gap on CAD/furniture is method-limited; on organic scans it is bounded by the singularity criterion; on room scans 90 % is out of reach at this face count. 2. **Lounge singularities.** QuadriFlow's integer stage fails on this TSDF room (inconsistent edge offsets in its subdivision), even after isotropic remeshing, so we fall back to Instant Meshes and inherit its singularity rate. 3. **Inverted / folded elements.** A few elements fold across features where the extracted layout crosses a crease diagonally, or where a recovered feature path cuts across the grid (visible as a sawtooth on fandisk): at most 0.8 % of faces, mean 0.34 % (QuadriFlow: up to 1.8 %, mean 0.47 %; Instant Meshes up to 1.1 %), hence the minimum SJ < 0 on some models. Untangling repairs most of them but not all; path recovery trades a few folds for exact feature loops. 4. **Thin parts.** Parts thinner than one element are represented approximately (single sheet) or by the decimated exact fallback, with locally irregular structure. 5. **Symmetry hint** assumes a near-symmetric input: the negative side is discarded. 6. **Density transitions** introduce valence-3/5 pairs along the refinement front (by design).

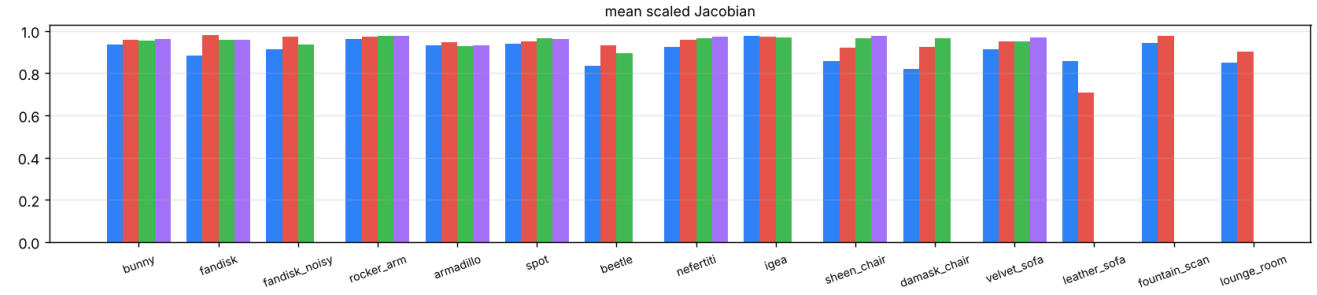
## 7. Future work

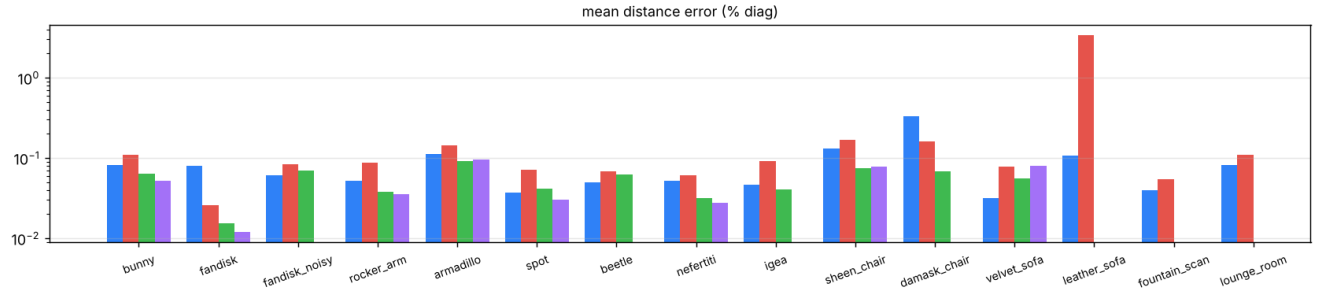
- Feature coverage: hard *integer* (position) constraints that put iso-lines exactly on creases in the integer-grid solve (QuadWild / QEx-style patch layouts), instead of recovering edges afterwards; anisotropic sizing (thin quads along creases) instead of the isotropic size band; transition templates for path subdivision that cost fewer irregular vertices than the current ~4 per split edge (the organic scans are limited by that cost).
- Replace the QuadriFlow extraction with a robust integer-grid map (QEx-style) to remove the fallback chain on scans.
- GPU field optimisation integrated in C++ (CUDA kernels for the orientation/position smoothing). Train the learned field on more data with a field-level loss that sees singularities, not only per-vertex angles.
- Quad-mesh valence-improvement operations (edge rotations, doublet/singlet removal, Bommes-style singularity pair cancellation) as a post-process.

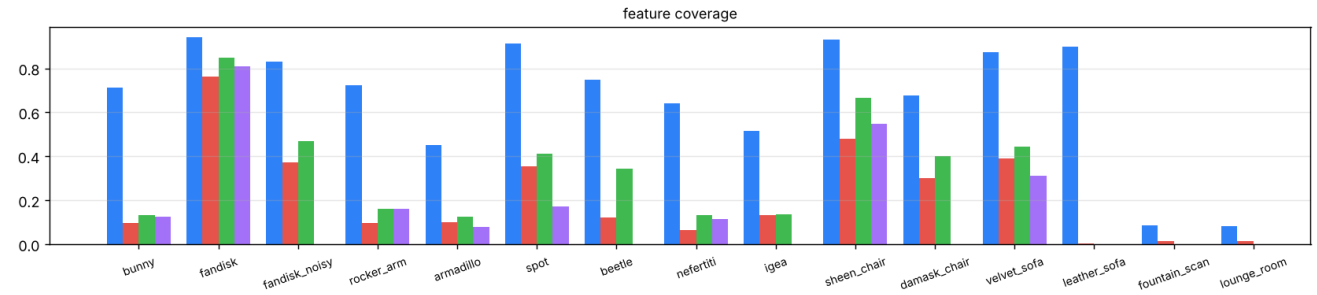
- Symmetry via field and integer constraints instead of clip-and-mirror, for approximately symmetric inputs.

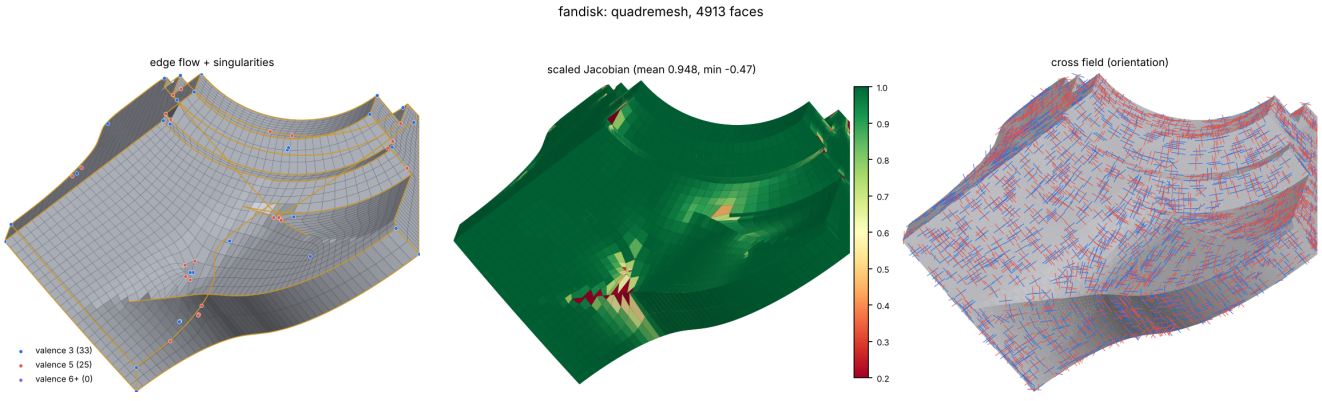
## Figures



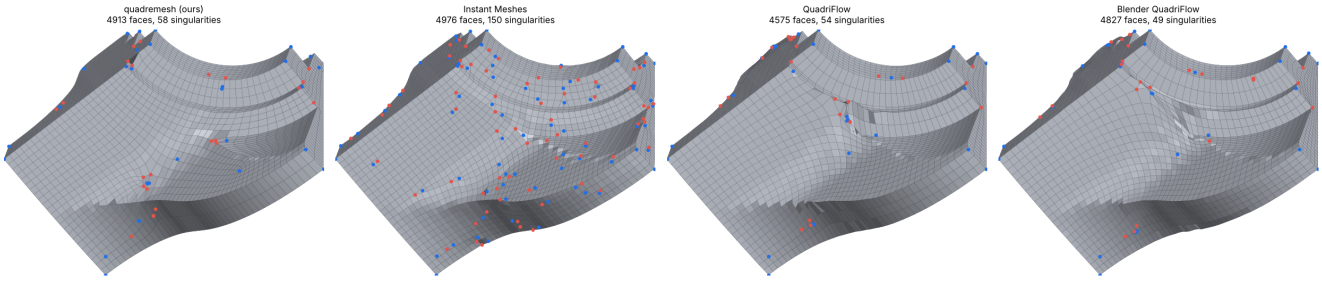



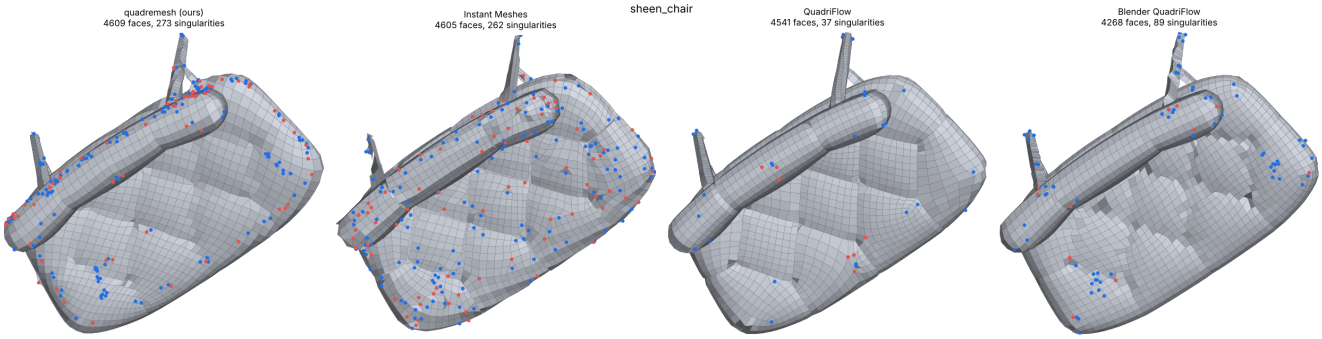



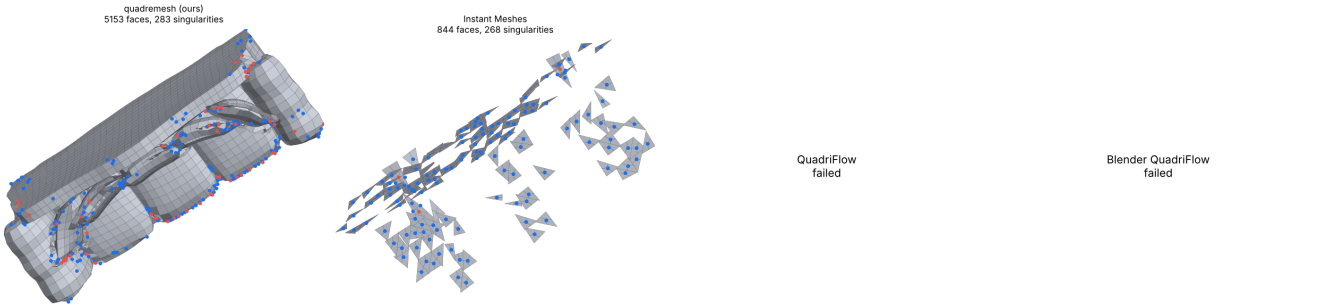
fandisk



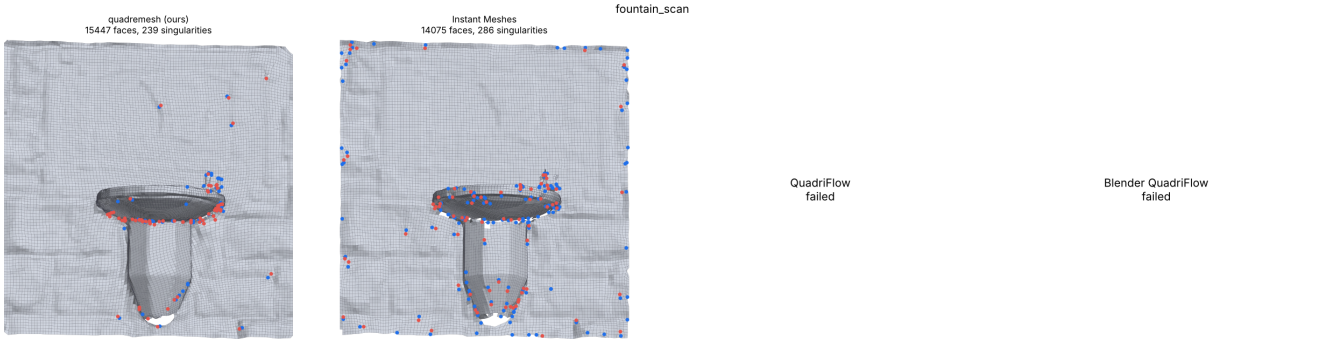

sheen_chair



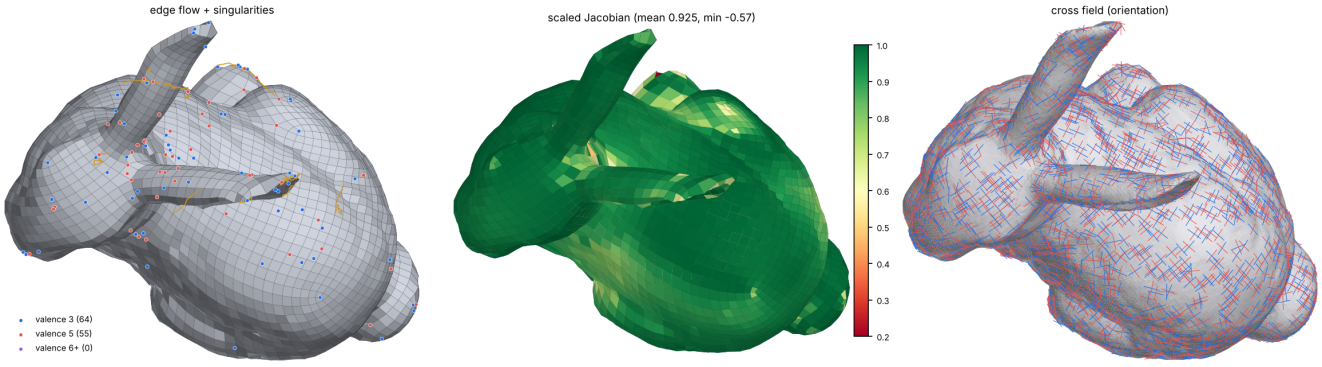

leather_sofa



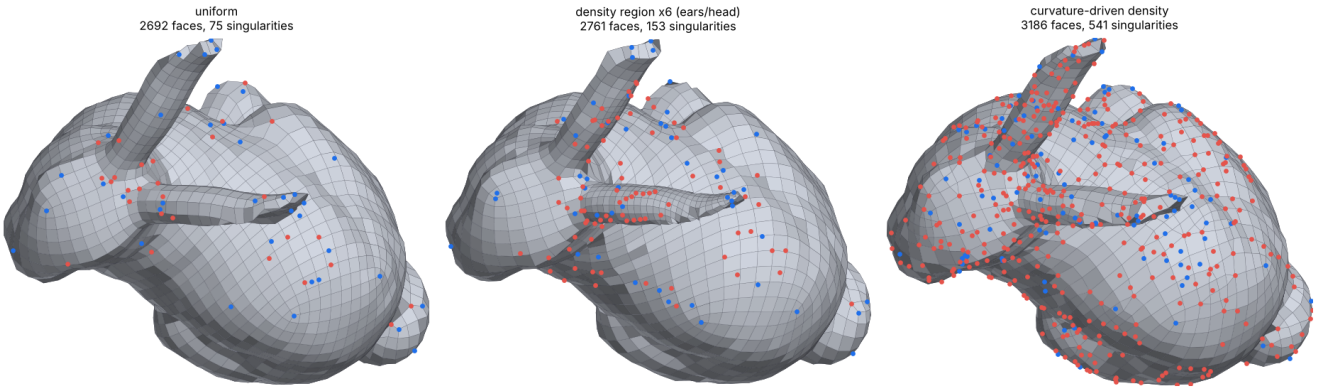

fountain_scan



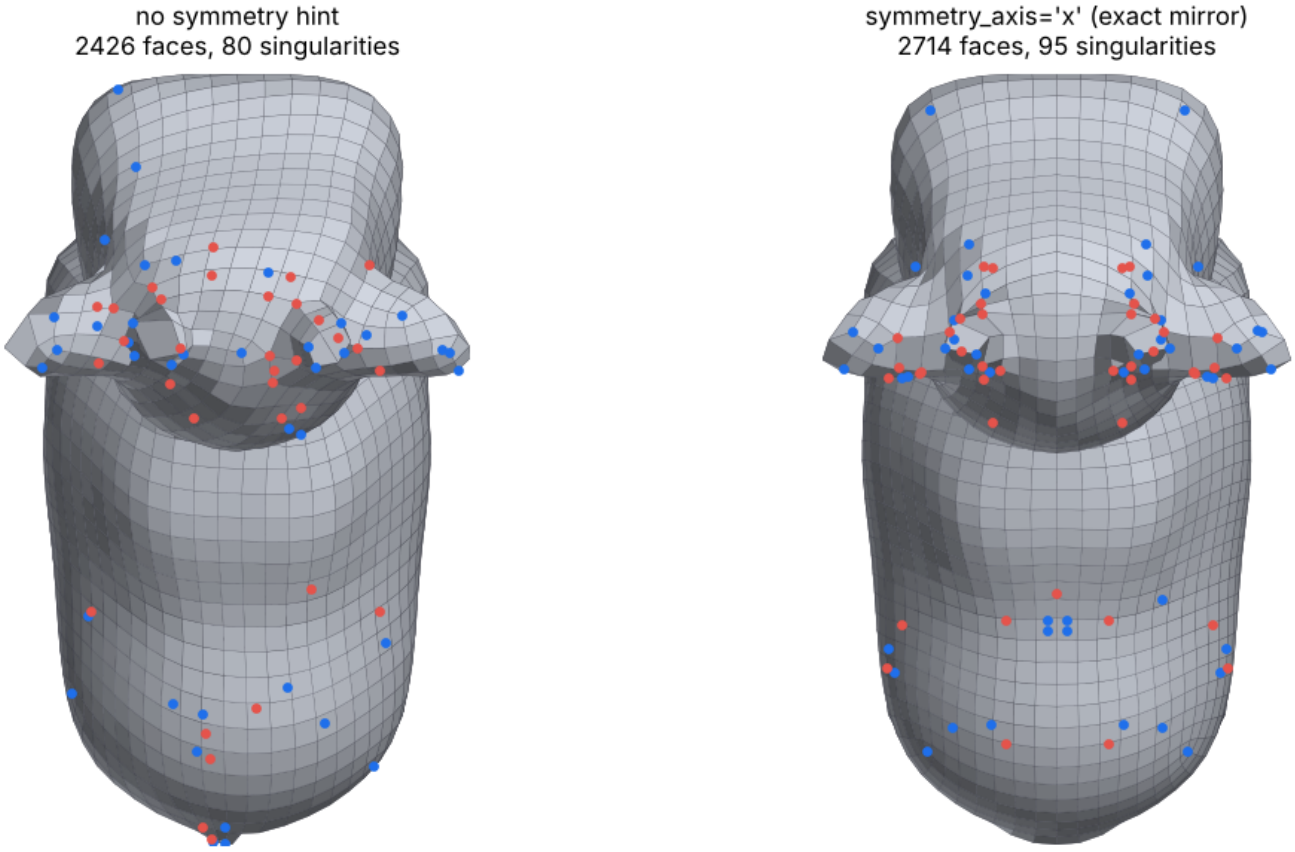

bunny: quadremesh, 4158 faces



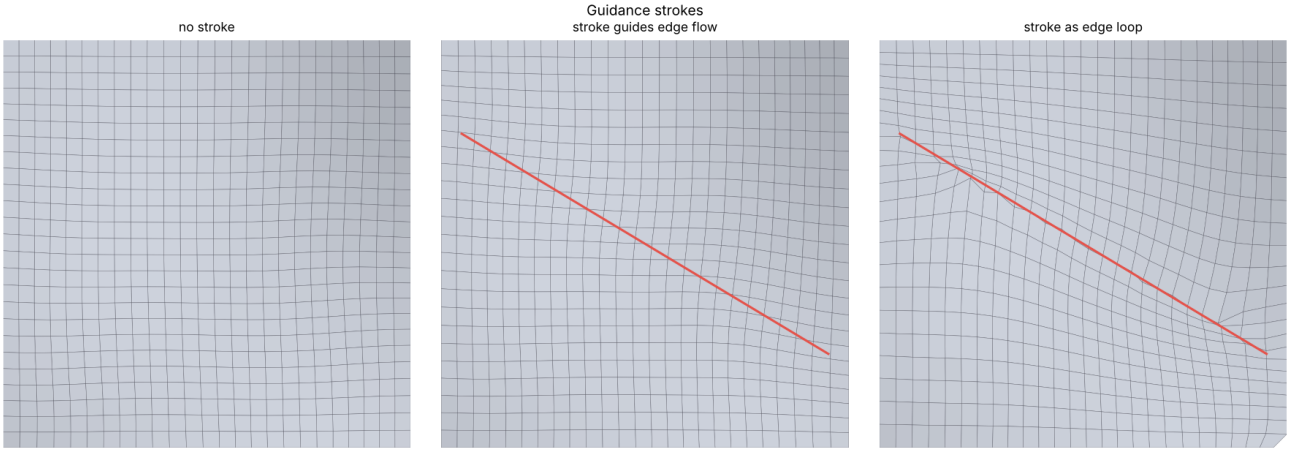
## Adaptive density map

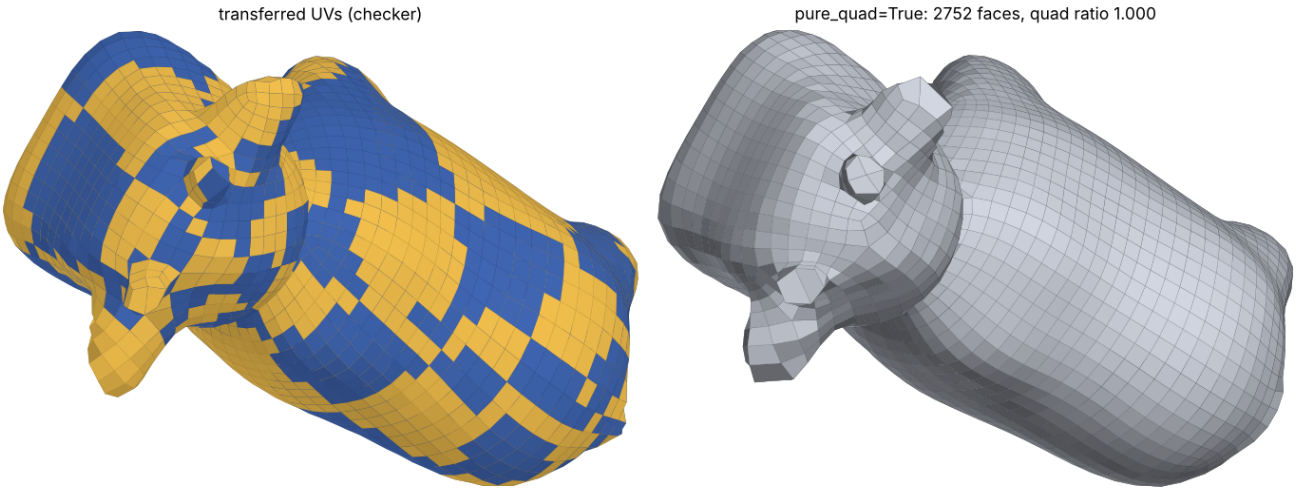



## Symmetry hint




## Guidance strokes






## 8. Reproducing

```
./scripts/fetch_deps.sh && cmake -S . -B build && cmake --build build -j
cp build/_quadremesh*.so python/quadremesh/
ctest --test-dir build && PYTHONPATH=python python -m pytest tests
python data/prepare_benchmark.py && python benchmark/run_benchmark.py
python benchmark/make_report.py && python benchmark/feature_demos.py
```

## 9. Code reference

Generated by `docs/gen_codemap.py` from the source: signatures and excerpts are copied from the files named.

### Libraries and what each one does here

| library | what it is | how it is used |
|---|---|---|
| Eigen 3 | header-only C++ linear algebra | Every mesh is stored as Eigen matrices (V: n×3 vertex positions, F: m×3 triangles, Q: k×4 quads); vectors, normals and the small least-squares systems in the field and relaxation code. |
| libigl | C++ geometry-processing library | AABB trees for closest-point queries (distance error, coverage test, projection onto the input surface, crease snapping) and the PLY reader. |
| QuadriFlow (vendored, 11 patches) | integer-grid quad solver, `third_party/qflow_patched` | Orientation field, position field, min-cost-flow singularity optimisation and quad extraction - the core solver. Patches: determinism, constraint bugs, external sharp edges / normals / scale, exceptions instead of exit(), bounded loops ( `PATCHES.md` ). |
| LEMON | graph library (inside QuadriFlow) | Network-simplex min-cost flow that QuadriFlow uses to move and cancel singularities globally. |
| Instant Meshes core | `imcore` , field-aligned remesher | Second extractor for parts QuadriFlow cannot solve (thin parts, the room scan), best-of-two on dense scans, and the internal reference for the singularity budget. |
| Intel TBB | parallel runtime (inside Instant Meshes) | Required by the Instant Meshes core; run single-threaded (nprocs = 1) for deterministic output. |
| Boost (headers) | C++ utilities | Required by LEMON's build. |
| cgltf | single-header glTF / GLB loader | Reads .gltf / .glb inputs (node transforms, UVs) in `src/io.cpp` . |
| pybind11 | C++ ↔ Python bindings | Builds the `_quadremesh` Python module from `python/bindings.cpp` ; every remesher parameter is a keyword argument. |

| library | what it is | how it is used |
|---|---|---|
| NumPy | Python arrays | Vertex / face arrays passed to and from the C++ core. |
| PyTorch | deep learning (optional component) | `python/quadremesh/learned.py` : a frame-equivariant graph network that predicts a guide field per vertex; runs on CUDA when available. |
| matplotlib, Pillow | plotting, images | `python/quadremesh/viz.py` and `benchmark/make_report.py` : the edge-flow / singularity / scaled-Jacobian / cross-field renders and the charts of the visual report. |
| trimesh, pymeshlab | Python mesh I/O and repair | Preparing the benchmark inputs ( `data/prepare_benchmark.py` , `scripts/prepare_lounge.py` ) and loading meshes in the Python scripts. |
| pynanoinstantmeshes, Blender | baselines | The Instant Meshes and Blender QuadriFlow runs of the benchmark ( `benchmark/run_benchmark.py` , `benchmark/blender_quadriflow.py` ). |
| markdown, WeasyPrint | documents | `docs/build_pdf.py` turns the write-up into the PDF. |

## Pipeline functions

### Load: `load_mesh`

- **File:** `src/io.cpp`
- **Input:** file path (.obj / .ply / .glb / .gltf / .off)
- **Output:** TriMesh: V (n×3), F (m×3), optional UVs
- **What it does:** Reads any supported format into one triangle mesh; glTF node transforms are applied.
- **Library:** cgltf, libigl

```
TriMesh load_mesh(const std::string& path)
```

### Preprocess: `preprocess`

- **File:** `src/preprocess.cpp`
- **Input:** raw V, F (+ per-vertex attributes), Params
- **Output:** WorkMesh: clean, manifold, consistently oriented triangles
- **What it does:** Welds duplicate vertices, removes degenerate faces, repairs non-manifold edges and vertices, orients faces, fills small holes, drops debris.
- **Library:** Eigen

```
WorkMesh preprocess(const MatrixXd& V, const MatrixXi& F, const MatrixXd& A, const Params& p,
PrepStats& st)
```

### Creases: `detect_features`

- **File:** `src/features.cpp`
- **Input:** V, F, angle threshold (35°), minimum chain length (3 L), support radius (L)
- **Output:** FeatureSet: sharp half-edges, crease segments, corner vertices
- **What it does:** Dihedral-angle test plus 1-ring and patch-scale tests, so scan noise is not a crease; chains shorter than 3 L are pruned.
- **Library:** Eigen

```
FeatureSet detect_features(const MatrixXd& V, const MatrixXi& F, double angle_deg, double min_len,
double support)
```

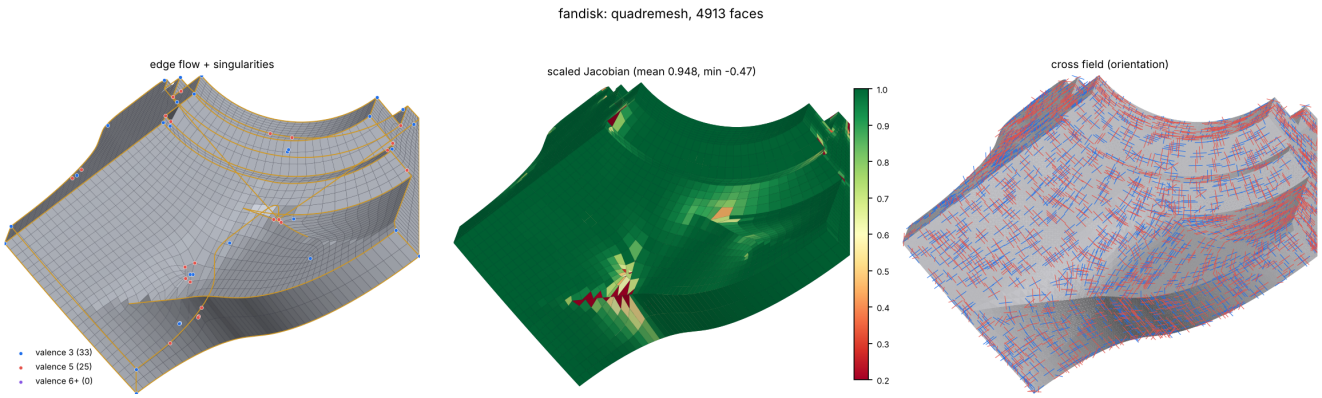



*Figure: `docs/report/fandisk_ours_panel.png` - creases (orange lines) and singularities on fandisk.*

**Field constraints: `build_constraints`**

- **File:** `src/field.cpp`
- **Input:** V, F, normals, crease flags, curvature, guide field, strokes
- **Output:** per-vertex direction + weight (hard on creases / boundaries, soft elsewhere)
- **What it does:** Turns creases, boundaries, curvature directions and the optional learned field into constraints for the cross field.
- **Library:** Eigen

```
FieldConstraints build_constraints(const MatrixXd& V, const MatrixXi& F, const MatrixXd& N,
                                   const std::vector<int>& he_sharp, bool boundary_hard,
                                   const CurvatureInfo* curv, double curvature_weight, const
MatrixXd* guide,
                                   const VectorXd* guide_w, double guide_strength,
                                   const std::vector<std::vector<Vector3d>>& strokes, double
stroke_radius,
                                   double soft_fade)
```

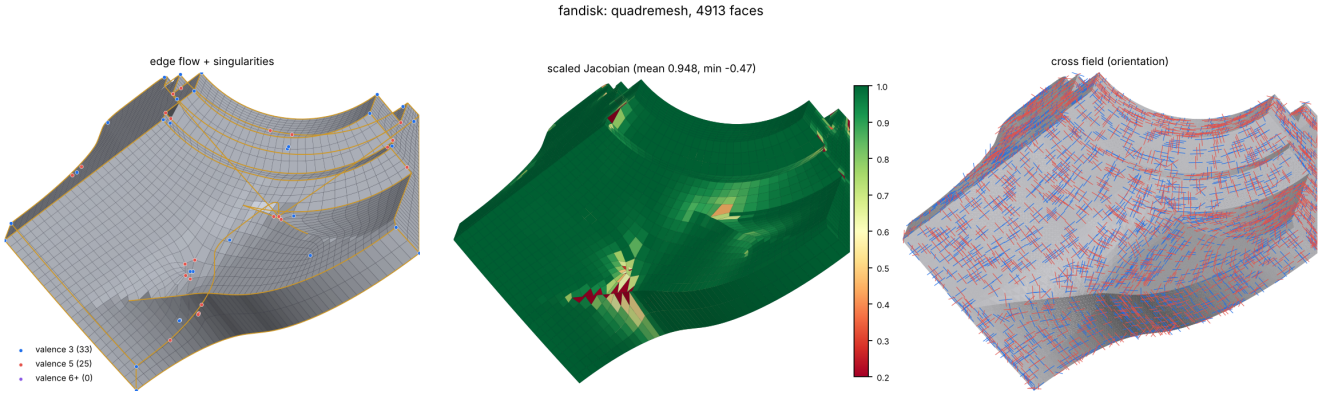



*Figure: `docs/report/fandisk_ours_panel.png` - right panel: the cross field.*

**Adaptive size: `density_to_scale`**

- **File:** `src/field.cpp`
- **Input:** V, F, per-vertex density, max ratio (8)
- **Output:** per-vertex scale target with the same expected face count
- **What it does:** Converts the crease-distance density into the scale the field solve aims for (S = d^-1/2, normalised).
- **Library:** Eigen

```
VectorXd density_to_scale(const MatrixXd& V, const MatrixXi& F, const VectorXd& density, double
max_ratio)
```

Excerpt ( `src/field.cpp` , line 218):

```
    // S = d^-1/2, normalised so that sum A/S^2 == sum A  (same expected quad count)
    VectorXd S(nv);
    double num = 0;
    for (int i = 0; i < nv; ++i) { S[i] = 1.0 / std::sqrt(d[i]); num += A[i] / (S[i] * S[i]); }
    double k = std::sqrt(num / std::max(at, 1e-30));
```

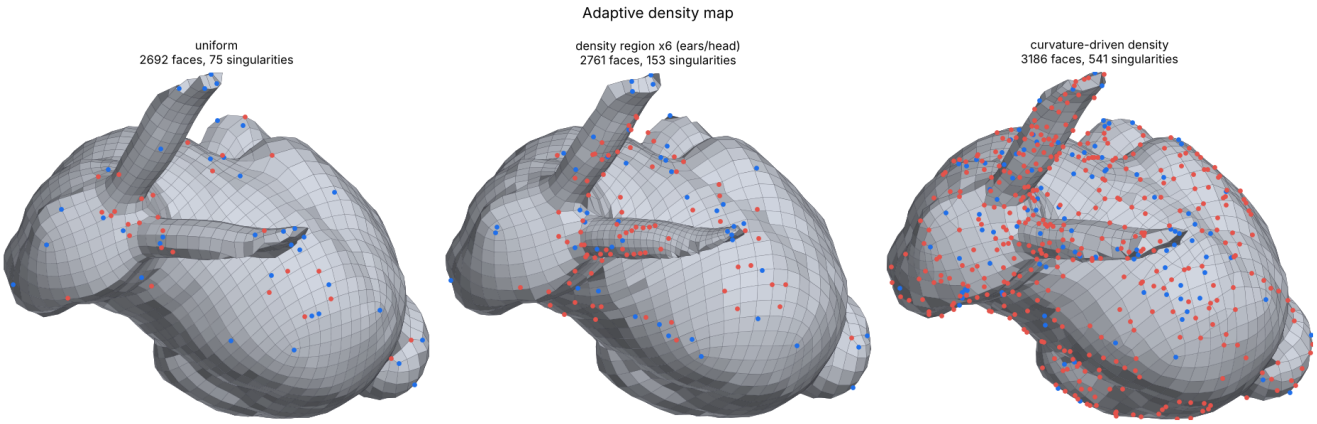



*Figure: `docs/report/demo_density.png` - adaptive density.*

**Integer grid: `run_quadriflow`**

- **File:** `src/qflow_driver.cpp`
- **Input:** one connected component: V, F, constraints, scale, face budget, seed
- **Output:** QFResult: quad vertices and faces, field for visualisation
- **What it does:** Runs the patched QuadriFlow: orientation field, scale, position field, min-cost-flow singularity optimisation, extraction.
- **Library:** QuadriFlow, LEMON

```
QFResult run_quadriflow(const QFJob& job)
```

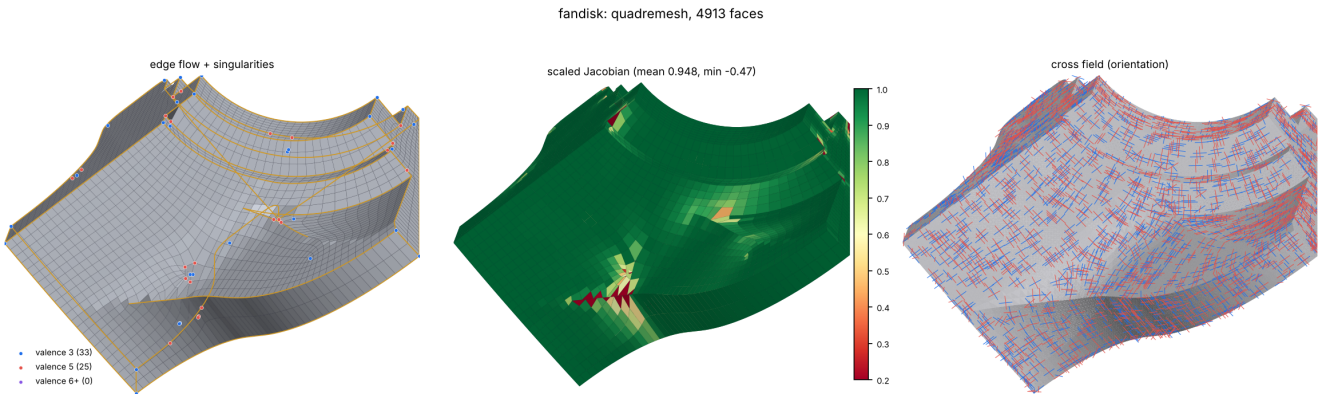



*Figure: `docs/report/fandisk_ours_panel.png` - left panel: the extracted quads.*

**Fallback extractor: `run_instant_meshes`**

- **File:** `src/im_driver.cpp`
- **Input:** V, F (flat arrays), face count, crease angle
- **Output:** IMResult: pure quads
- **What it does:** Instant Meshes extraction for parts QuadriFlow cannot handle, and the reference for the singularity budget.
- **Library:** Instant Meshes core, TBB

```
IMResult run_instant_meshes(const std::vector<double>& Vin, const std::vector<int>& Fin, int faces,
                            double crease_angle_deg, bool align_boundaries)
```

**Whole pipeline:** `quadremesh`

- **File:** `src/remesh.cpp`
- **Input:** input TriMesh + Params (target faces, feature angle, options)
- **Output:** Result: quad mesh, field, crease segments, log, timings
- **What it does:** Runs every stage per component, the coverage stage, the quality guard, face-count control and (optionally) the coverage search.
- **Library:** all of the above

```
Result quadremesh(const TriMesh& input, const Params& p)
```




*Figure:* `docs/report/charts.png` *- per-method metrics across the benchmark.*

**Crease paths + relaxation:** `relax_and_project`

- **File:** `src/postprocess.cpp`
- **Input:** quad mesh, reference surface with creases, L, iterations
- **Output:** same mesh, vertices moved: edge paths snapped onto creases, smoothed, reprojected

- **What it does:** Directed-edge Dijkstra finds an edge path along every crease and snaps it on; then tangential relaxation and reprojection onto the input surface.
- **Library:** libigl (AABB)

```
void relax_and_project(QuadMesh& m, const SurfaceRef& ref, double L, int iterations, PostStats& st)
```

Excerpt ( `src/postprocess.cpp` , line 640):

```
                double c = len * (1.0 + 6.0 * dm / L) + len * 1.5 * (1.0 - align);
                if (pu >= 0 && corner[u].count({std::min(pu, v), std::max(pu, v)})) c += 3.0 * L;
```

**Near-miss snapping:** `snap_near_creases`

- **File:** `src/crease_cut.cpp`
- **Input:** quad mesh, crease polylines, L, max distance (0.5 L), max angle (75°)
- **Output:** number of edges snapped; mesh vertices moved onto creases
- **What it does:** Moves both ends of an edge that runs beside an uncovered crease onto it, if quality and existing coverage hold.
- **Library:** libigl (AABB), Eigen

```
int snap_near_creases(QuadMesh& m, const std::vector<std::vector<Vector3d>>& chains, double L,
double max_dist,
                      std::vector<Vector3d>* moved, double max_angle)
```

Excerpt ( `src/crease_cut.cpp` , line 727):

```
        Vector3d qa = closest_on_chain(S[i].c, pa), qb = closest_on_chain(S[i].c, pb);
```

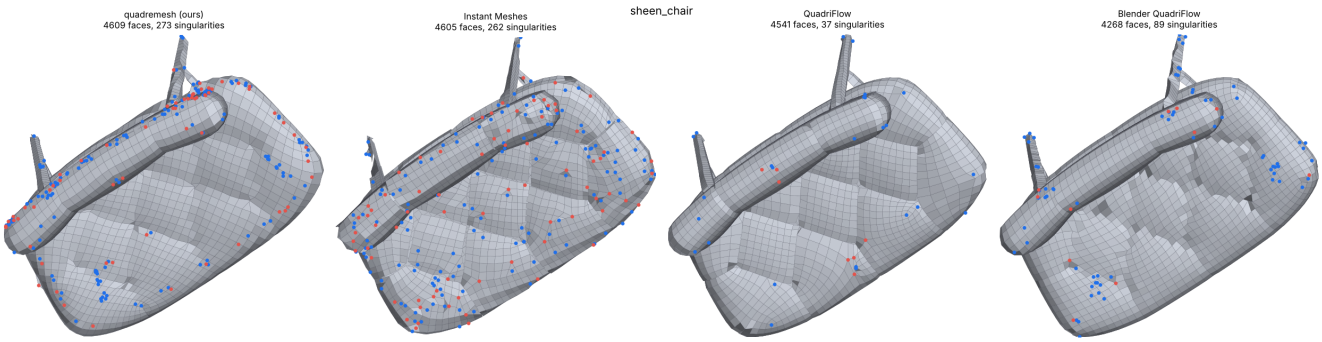



*Figure: `docs/report/sheen_chair_compare.png` - edge flow along the chair's piping.*

**Path-edge subdivision:** `subdivide_path_edges`

- **File:** `src/crease_cut.cpp`
- **Input:** quad mesh, crease edge paths, crease polylines, L, pieces (3), max edges
- **Output:** number of edges split; mesh with new crease vertices, still all quads
- **What it does:** Splits path edges whose chord misses a curved crease into three at crease points; each neighbouring quad becomes two quads.
- **Library:** Eigen

```
int subdivide_path_edges(QuadMesh& m, const std::vector<std::vector<int>>& paths,
                         const std::vector<std::vector<Vector3d>>& chains, double L, int pieces, int
max_edges,
                         std::vector<Vector3d>* added)
```

Excerpt ( `src/crease_cut.cpp` , line 926):

```
            // a quad that became a hexagon (a n1 n2 b c d) is split right here, with the diagonal
            // from n1 on one side and from n2 on the other, so no new vertex is left with valence 2
```

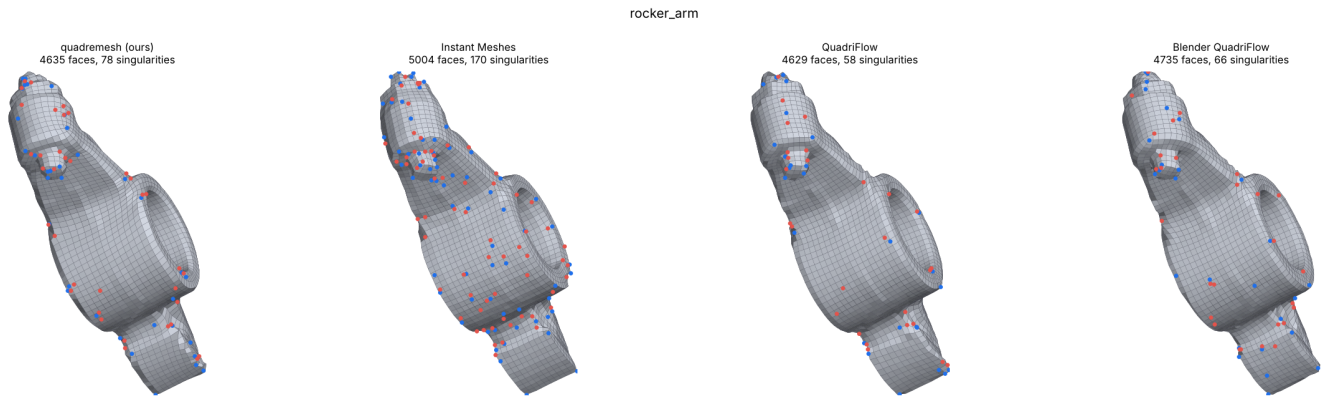



*Figure: `docs/report/rocker_arm_compare.png` - curved creases on the rocker arm.*

**Coverage check: `uncovered_crease_runs`**

- **File:** `src/crease_cut.cpp`
- **Input:** quad mesh, crease polylines, L, tolerances (0.1 L, 25°)
- **Output:** the crease pieces not yet covered (+ covered / total length)
- **What it does:** The evaluator's coverage test, used inside the pipeline to decide what to fix.
- **Library:** libigl (AABB)

```
std::vector<std::vector<Vector3d>> uncovered_crease_runs(const QuadMesh& m,
                                                         const std::vector<std::vector<Vector3d>>&
chains, double L,
                                                         double min_run, double pad, double tol,
double max_angle,
                                                         double* covered, double* total)
```

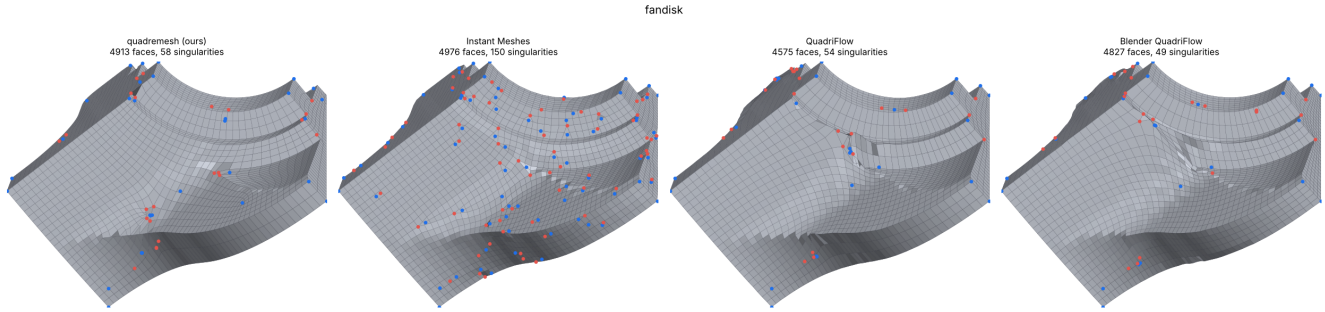



*Figure: `docs/report/fandisk_compare.png` - fandisk, all methods.*

**Scaled Jacobian: `scaled_jacobian`**

- **File:** `src/metrics.cpp`
- **Input:** vertex positions, the 4 corner indices of a quad
- **Output:** number in [-1, 1]: worst corner's signed sine
- **What it does:** Shape quality of one quad; used by the evaluator, the quality guards and the snapping checks.
- **Library:** Eigen

```
double scaled_jacobian(const MatrixXd& V, int a, int b, int c, int d)
```

Excerpt ( `src/metrics.cpp` , line 30):

```
    int q[4] = {a, b, c, d};
    Vector3d n = Vector3d::Zero();
    for (int k = 0; k < 4; ++k) {
        Vector3d e1 = V.row(q[(k + 1) % 4]) - V.row(q[k]), e2 = V.row(q[(k + 3) % 4]) - V.row(q[k]);
        n += e1.cross(e2);
    }
    double ln = n.norm();
    if (ln <= 0) return -1;
    n /= ln;
    double m = 1e9;
    for (int k = 0; k < 4; ++k) {
        Vector3d e1 = V.row(q[(k + 1) % 4]) - V.row(q[k]), e2 = V.row(q[(k + 3) % 4]) - V.row(q[k]);
        double den = e1.norm() * e2.norm();
        if (den <= 0) return -1;
        m = std::min(m, e1.cross(e2).dot(n) / den);
    }
    return m;
}
```

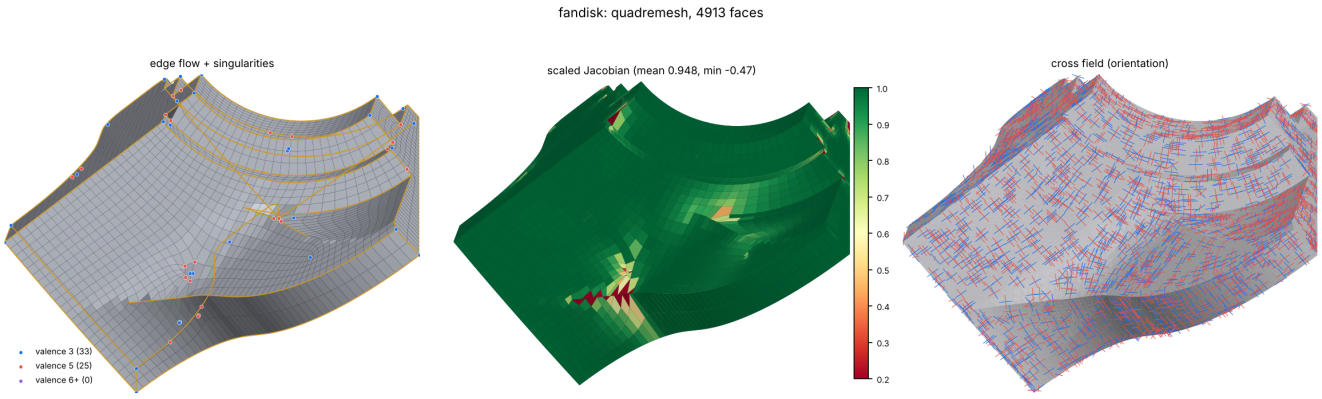



*Figure: `docs/report/fandisk_ours_panel.png` - middle panel: scaled-Jacobian heatmap.*

**Evaluator: `evaluate`**

- **File:** `src/metrics.cpp`
- **Input:** input triangles, output quads, EvalParams (angle 35°, target faces)
- **Output:** JSON report: quad ratio, SJ, distance, feature coverage, singularities, topology
- **What it does:** The one scoring function used by the CLI report, quadeval, the benchmark and the coverage search.
- **Library:** libigl (AABB), Eigen

```
std::string evaluate(const TriMesh& input, const QuadMesh& out, const EvalParams& ep, double
runtime, const std::string& method)
```

Excerpt ( `src/metrics.cpp` , line 367):

```
        J.num("feature_coverage", ft > 0 ? fc / ft : 1.0);
```

**UV transfer: `transfer_uvs`**

- **File:** `src/transfer.cpp`
- **Input:** input mesh with UVs, output quads
- **Output:** output quads with per-corner UVs
- **What it does:** Each quad samples the UV chart under its centroid, so faces never straddle a seam.
- **Library:** libigl (AABB)

```
void transfer_uvs(const TriMesh& src, QuadMesh& out)
```

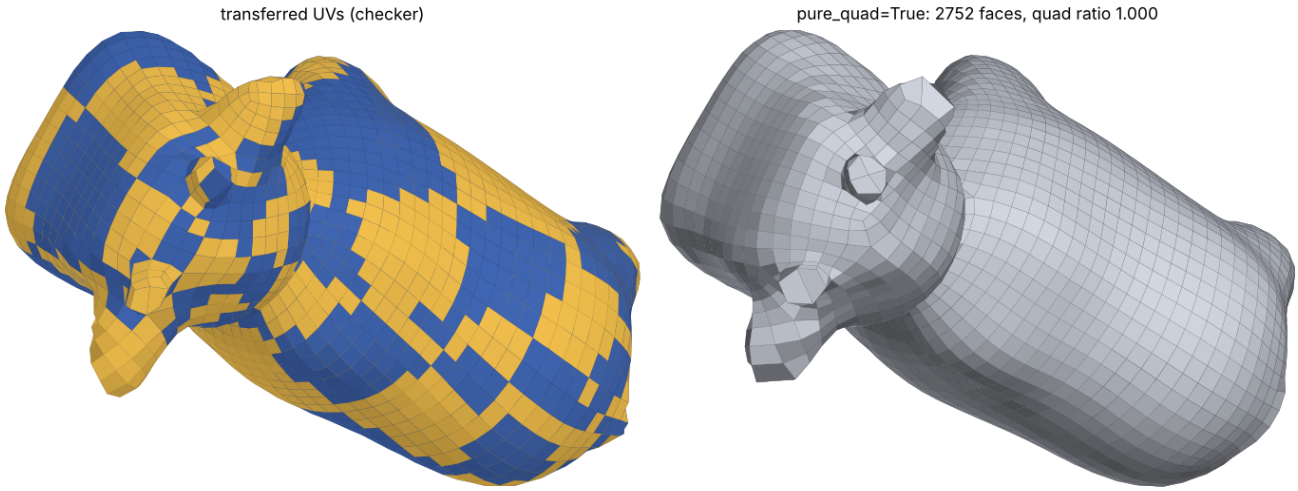



*Figure: `docs/report/demo_uv_purequad.png` - UV transfer and pure-quad output.*

**Python API: `quadremesh`**

- **File:** `python/quadremesh/api.py`
- **Input:** path / (V, F) / mesh object, target_faces, feature_angle, options
- **Output:** QuadRemeshResult: V, Q, UV, quality report
- **What it does:** Python front end; keyword options map one-to-one to the C++ Params.
- **Library:** pybind11, NumPy

```
def quadremesh(input: MeshLike, target_faces: int = 5000, feature_angle: float = 35.0, *,
               output: Optional[str] = None, report: Union[bool, str, None] = True,
               target_edge_length: Optional[float] = None, strokes: Optional[Sequence[np.ndarray]] =
None,
               strokes_as_edges: bool = False, density: Optional[np.ndarray] = None,
               density_regions: Optional[Sequence[Tuple[Sequence[float], float, float]]] = None,
               curvature_density: float = 0.0, symmetry_axis: Union[str, int, None] = None,
```

**Learned field: `predict_guide_field`**

- **File:** `python/quadremesh/learned.py`
- **Input:** V, F, checkpoint, device
- **Output:** per-vertex guide direction + confidence
- **What it does:** Optional PyTorch component: predicts a guide field that the remesher uses as soft guidance; CUDA when available.
- **Library:** PyTorch

```
def predict_guide_field(V, F, checkpoint: Optional[str] = None, device="auto", feature_angle=35.0):
```